

More Than Shapes: Exploring the Tactile Parameters of Art Appreciation for the Visually Impaired

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ABSTRACT

Visual art appreciation for the visually impaired is not only a personal need, but also a need for equal social development. Our study investigates how tactile sensations can compensate for visual impairments in art appreciation. We assembled a focus group of experts to translate visual art elements into tactile equivalents by applying theories of art knowledge and appreciation. We deconstructed paintings into five primary elements—composition, content, color, light and shadow, brushstroke—and two secondary elements—style and emotion. These elements were then converted into tactile paintings for a series of Tactile Aesthetic Workshops aimed at visually impaired participants. The results showed that these tactile representations effectively allowed visually impaired individuals to appreciate art, with a success rate of **78.27%**. Moving forward, we plan to enhance this approach by integrating olfactory and auditory elements to create a more immersive augmented reality (AR) art appreciation environment for the visually impaired.

CCS CONCEPTS

• **Human-centered computing**→**Human computer interaction (HCI)**→**Interaction devices**→**Haptic devices**;

KEYWORDS

Tactile interaction, Visually Impaired, Multimodal, Art Appreciation

ACM Reference Format:

MingYu Cui, Chao Yuan, Yujia Liu, and Yingying Zheng. 2024. More Than Shapes: Exploring the Tactile Parameters of Art Appreciation for the Visually Impaired. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678391>

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
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ACM ISBN 979-8-4007-1058-2/24/10...\$15.00
<https://doi.org/10.1145/3675094.3678391>

1 INTRODUCTION

According to the World Health Organization (WHO), over 250 million people globally are visually impaired, with 37 million being completely blind [18]. While visual art remains largely inaccessible to them, art appreciation, a right deeply rooted in Schiller's 1974 "Briefs on Aesthetic Education" [1], is crucial for recognizing and experiencing beauty. This capacity is as vital for the visually impaired as for anyone else.

Traditional methods for aiding visually impaired individuals "see" artworks, such as transforming paintings into tactile shape sculptures, often oversimplify the art, capturing only the basic essence of subjects like portraits and still-lives. Multimodal approaches involving scents, music, or verbal explanations add emotional depth but lack the exploratory and aesthetic immersion needed.

To overcome these limitations, we introduced serialized tactile elements. Firstly, we conducted a focus group discussion to explore how we can systematically analyse the knowledge of painting and present the elements of painting in a multi-dimensional way. Secondly, we deconstructed the five **Primary elements** of painting: **Layout**, **Content**, **Colour**, **Lighting**, and **Brushwork** to express the two **Secondary elements**: **Style** and **Emotion**. We transformed these painting elements into serialised tactile elements and made tactile banners to display visual artworks in multiple dimensions for the visually impaired.

Finally, we conducted a tactile art workshop where visually impaired participants could engage with these artworks. A subsequent Q&A session assessed the effectiveness of this tactile appreciation, achieving a **78.27%** accuracy rate in line with expert evaluations. User feedback has been overwhelmingly positive; one participant remarked, "I can feel the painting myself, gaining my own understanding rather than relying on a simplified explanation." This indicates that serialized tactile elements can effectively substitute for vision, allowing visually impaired individuals to independently appreciate and interpret art.

2 RELATED WORK

Research on enabling visually impaired individuals to "see" art has shown their sharper sense of touch than those who can see [4, 6]. Tactile research has led to innovations [7, 15, 19, 25, 26], like mechanical dots for basic shapes [13] and modular drawing tools [20],

enhancing tactile art representations [16]. In art appreciation, multi-channel AR techniques—using music [9, 28], sonified guidance [8], and touchable VR [22] aid visually impaired people. Furthermore, targeted acoustic games have been developed for this audience [10]. Educationally, 3D printing is utilized to create tactile relief sculptures and translate colors into textures, improving sensory learning for the visually impaired [2, 11]. However, these studies face two main issues: 1. Haptic methods often oversimplify complex artworks into basic shapes. 2. Multimodal techniques, though rich in sensory detail, may neglect the deeper exploratory and aesthetic engagement needed for comprehensive art appreciation by visually impaired users.

3 ELEMENTS OF ART APPRECIATION

3.1 Panel of Experts

In order to systematically analyse the knowledge of painting and present the elements of painting in a multi-dimensional way, we obtain painting elements and aesthetic order through experts. We invited a focus group of nine experts, each with over five years of professional experience ($M=11.78$), including three from painting, three from art theory, and three from art history. During the focus group discussion, the experts chose six classical aesthetic approaches to painting [12, 14, 23] as the theoretical basis from which the appreciation elements were extracted and graded:

- 1 Layout, Content, Color, Lighting, and Brushwork→Visual direct expression→**Primary elements**
- 2 Style, emotion→Processing informations→**Secondary elements**, as shown in Figure 1

Theory	Author	Elements&classifications	
Six methods	Xiehe	Brushwork, Content, Color, Layout, Technique	Vitality
Appreciation the three-stage model	Erwin Panofsky	Vor-Ikonographie Color, Lighting, Line, Brushwork, Perspective, Composition, Layout	Ikonographie Symbols, Emotion, Atmospheres, Style, History
Iconography	Ancient Greece	Natural objects Image, Lines, Color, Shape, Content	Social images Style, Intention, Social, Cultural, Historical
Formal Analysis	Roger Fry	Formal Structure, Brushwork, Lines, Shape	Rhythm Style, Emotion
Planness of modernism	Clement Greenberg	Performance Light and shadow, space, color, Layout	Symbol
Observe the theory of penetration	Hanson	Accept information Solve the space, Perspective, Color, Strong and weak rhythm	Processing information
		Visual direct expression→ Primary elements	Processing information→ Secondary elements

Figure 1: Explanation of the theories of classical painting appreciation and their Elements discussed by the panel of experts.

3.2 Artwork Analysis

The panel chose to analyze *Impressionist painting*, a significant art movement that originated in 1860s France. Known for its dynamic compositions and bright colors, Impressionism emphasizes sensory impressions of light and shadow, focusing on pure visual experience. While not easily translatable into traditional sculptures, it allows visually impaired people to achieve an aesthetic experience through tactile exploration, aligning well with this research..

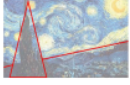
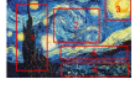
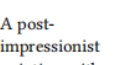
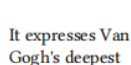
Name	Layout	Content	Brushwork
			
<i>The Starry Night</i>	7:3 up and down. The main object is located on the left side.	1. Trees 2. Moving air 3. Stars and moon 4. Villages 5. Mountains.	Thick and short brushstrokes are woven into curving, swirling lines.
Color	Light	Style	Emotion
			
Strong yellow and blue color contrast.	Back lighting, the sky is bright above and main object are dark.	A post-impressionist painting with unadorned brushstrokes and an emphasis on color and light.	It expresses Van Gogh's deepest loneliness, restlessness and struggle, hopes and dreams.

Figure 2: Interpretive graphical representation of serialised elements, exemplified by the starry night.

We selected 10 classic *Impressionist painting* to analyse each element. (image credit: Wiki-Art at <https://b.yishu.wiki>). We use *Van Gogh's The Starry Night* as an example to show that it is a post-impressionist work, as shown in Figure 2.

4 TRANSFORM TO TACTILE

The focus group, consisting of researchers and experts, aims to develop an effective method for converting visual elements into tactile elements and determining appropriate tactile perception parameters. We present the primary serialized elements identified by the expert group in a hierarchical structure and use 3D printing to create tactile surfaces for evaluation by visually impaired individuals.

Layout→The image is segmented into large geometric areas with distinct height differences at the edges of adjacent geometries. Main objects are abstracted into shapes such as circles, triangles, and squares, presented as geometric protrusions. **Content**→Lines outline the content, with the degree of protrusion indicating perspective depth. Lines with deeper protrusions represent foreground objects, while shallower protrusions represent background objects. **Color**→The artwork is divided into 3 to 4 primary color systems, with the depth of protrusions reflecting color intensity. **Lighting**→The artwork is processed into illuminated and shadowed parts, with bright areas protruding and dark areas recessed. **Brushwork**→The primary brushstrokes are represented as individual protrusions.

The basic technical methods of the transformation is shown in Table 1. We transformed 10 classic Impressionist paintings, using "Impression, Sunrise" as an example, as shown in Figure 3.

5 TACTILE AESTHETICS TESTING

The researchers invited 20 visually impaired students from Beijing Union University to participate in the study. Additionally, 10 art and design students volunteered to provide accessibility services to the visually impaired participants., as shown in Figure 4.

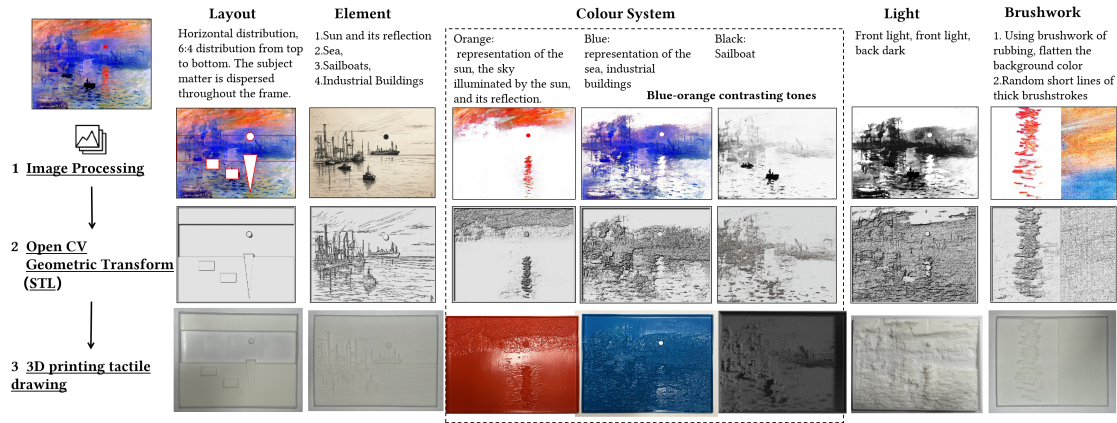


Figure 3: Steps for converting serialized elements of "Impression, Sunrise" into tactile relief images and their presentation

Theory	Steps	Formality
Layout	Segmentation based on the main color block, while marking the location of the main object	Manually
Content	Access to AI drawing interface, access to Midjourney Api key, recognition of main content and retention of line drawings	AI
Color	Identifies specific colors, using CV in range to preserve a certain range of colors. <code>lower_blue = np.array([100, 150, 50])</code> <code>upper_blue = np.array([140, 255, 200])</code>	Open CV
Light	Increase the contrast between light and dark by adjusting the color contrast of the image. Keep the light channel, dark channel, remove the middle color channel	Open CV
Brushwork	Identify the main strokes in the picture, extract the upper strokes, zoom in locally and enhance the contrasts.	Manually + Open CV
Conversion to 3D	Using grey level transformations, geometric transformations are done	Open CV

Table 1: Optimized table ensuring content fits within the page width

5.1 Participants

We first conducted in-depth interviews with the blind students. We investigated their basic information: 1.Level of visual impairment, 2.Visual acuity, 3.Range of visual field, 4.Basic conditions (Congenitally blindness, Acquired blindness, Normal visual field with weak visual acuity, Normal visual acuity with abnormal visual field), 5.Knowledge of colors, 6.Experience in painting or handicrafts.

Fifteen participants had prior experience in crafts and painting and were eager to appreciate visual artworks. They used screen readers on mobile phones to learn about painting in their daily lives. Remarkably, they developed their own methods of understanding colors through daily practice, enabling them to match clothes and select colors for objects in their lives. [3, 17].

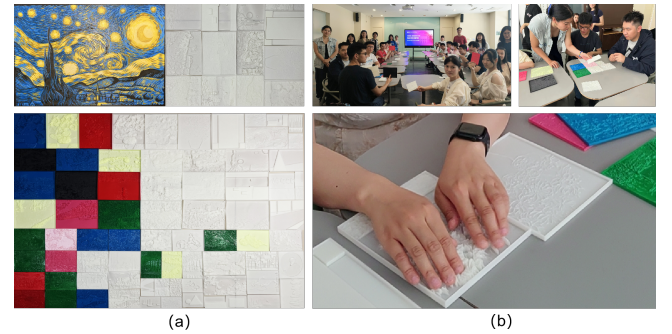


Figure 4: Tactile Aesthetic Education Workshop for the Visually Impaired: (a) Material preparation, tactile relief shown by Level 1 serialised painting elements. (b) Visually impaired students appreciating artworks through touch.

5.2 Preparation

Researchers first conducted a lecture on Impressionist painting for the visually impaired students, covering aspects such as composition, content, color, lighting, and brushwork. They then randomly selected five paintings for exploratory learning with the students. On average, each tactile surface took 4 minutes and 35 seconds to explore, with the total perception time for all paintings being 30 minutes.

5.3 Testing Process

We chose five different categories of paintings for visually impaired students to perceive independently, including landscapes, portraits and scenes with both landscapes and figures. In addition, we printed two Chinese paintings and two classical oil paintings for visually impaired users to judge their styles. Based on discussions with the panel, we designed an appreciation questionnaire. It consisted of 26 questions: $5 \times 5 = 25$ questions related to the main sequence elements (see Figure 6) and 1 question for style judgement. Volunteers ask visually impaired students questions about their perceptual results in a question and answer format. A total of 20 users answered 520 questions with an overall accuracy of 78.27%.

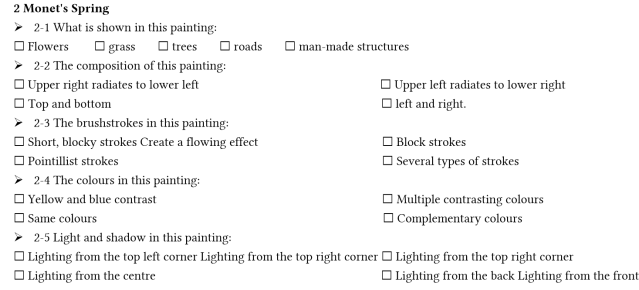


Figure 5: Problem setting: Monet's Spring (the second painting) as an example.

5.4 Data analysis

5.4.1 Elements correction. In the testing of elements, layout, content, and style received above-average scores, whereas color, lighting, and brush strokes scored below average, with lighting receiving the lowest score. Regarding the element scores, we queried users for explanations. For instance, concerning lighting, 10 users indicated that the content and knowledge of lighting itself were difficult to comprehend, leading to misjudgments; additionally, five users mentioned the difficulty in perceiving this element through touch. It was challenging to determine whether the concave-convex effect was due to dark colors or backlighting. The results and analysis of the elements are illustrated in **Figure 6**.

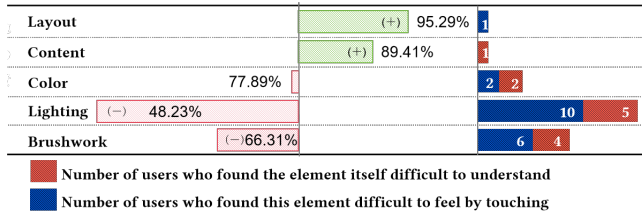


Figure 6: Elemental results analysis and data feedback.

5.4.2 Hotograph correction. Analysis of the accuracy of elements corresponding to each painting was conducted, with results depicted in **Figure 7**. The brushstrokes in the first painting "Pear" were challenging to comprehend, while the S-shaped lighting in the third painting "Poplar Tree" posed difficulties in understanding. Variations in scores were observed across paintings, with landscapes and still-life paintings exhibiting higher accuracy rates. Visually impaired users encountered challenges in tactile exploration of artwork rich in tactile content, such as the fourth painting "Summer Afternoon on the Dabang Island."

5.4.3 Type of visual impairment. When categorized by type, results showed that some users with partial vision performed notably better than the other two categories. Variances between congenitally blind and adventitiously blind individuals were minimal, with user interviews suggesting that the latter group possessed superior art exposure compared to the former, albeit with relatively weaker tactile perceptual abilities, resulting in nearly identical outcomes. in **Figure 8**. Subsequent user experiments could involve setting up control groups to delve deeper into these findings.

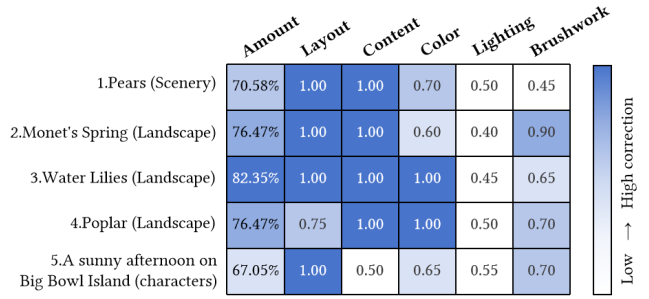


Figure 7: Statistical results categorised by pictures.

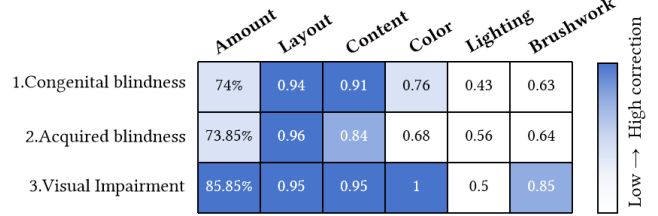


Figure 8: Statistical results categorised by Type of visual impairment.

5.4.4 Secondary element. Through tactile experiences, visually impaired individuals demonstrated the ability to discern various painting styles and make accurate judgments through touch ($P=82.35$). They shared their perceived painting styles and emotional interpretations of the artists with volunteers, enriching their subjective experiences.

6 DISCUSSION AND FUTURE WORK

The conclusions and future work that the current phase of the research provides for the design of AR artwork appreciation are:

1. The serialization display (3.1) and comparative display methods (5.2) of five basic elements will be automated in subsequent studies in the form of mechanical matrices at different temperatures [21, 27].

2. The appreciation of elements that are difficult to understand and perceive through touch, such as lighting and brushstrokes, requires accompanying knowledge explanations through sound channels (5.4.1).

3. In this study, we used tactile surfaces to represent the secondary elements of style and emotion(5.4.4). Previous research has matched these elements with music, scent, and tactile vibrations[5, 24]. Future studies will focus more on these secondary elements and aim to integrate multimodal interactions, with an emphasis on touch.

Tactile perception serves as the primary channel for visually impaired individuals to appreciate visual artworks in AR experiences. In future research, we will enrich tactile experiences by improving materials, temperatures, and other factors. Additionally, we will continue to explore auditory and olfactory aspects related to secondary elements, providing users with immersive multimodal AR art appreciation experiences.

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